Math 510 **Fall 2026**
Study Sheet for Quiz 4 **Drew Armstrong**

- Let $f : U \to V$ be a linear function. The image is $im(f) = \{f(\mathbf{u}) : \mathbf{u} \in U\}$ and the kernel is $ker(f) = \{\mathbf{u} \in U : f(\mathbf{u}) = \mathbf{0}\}$. We have f surjective $\Leftrightarrow$ $im(f) = V$ and f injective $\Leftrightarrow$ $ker(f) = \{\mathbf{0}\}$.

- Let $f : U \to V$ be linear. The *rank-nullity theorem* says
$$\boxed{\dim im(f) + \dim ker(f) = \dim U}.$$
It follows from this that f injective implies $\dim U \leq \dim V$, f surjective implies $\dim U \geq \dim V$, and f bijective (invertible) implies $\dim U = \dim V$. If $\dim U = \dim V$, then $f : U \to V$ is injective if and only if it is surjective.

- A matrix $A \in \mathbb{C}^{m \times n}$ with m rows and n columns is the same as a linear function $A : \mathbb{C}^n \to \mathbb{C}^m$. The image is the column space $\mathcal{C}(A) = \{A\mathbf{x} : \mathbf{x} \in \mathbb{C}^n\}$ and the kernel is the nullspace $\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{C}^n : A\mathbf{x} = \mathbf{0}\}$. Hence rank-nullity says
$$\boxed{\dim \mathcal{C}(A) + \dim \mathcal{N}(A) = \dim \mathbb{C}^n = \#\text{ columns of } A}.$$

- If a matrix $A \in \mathbb{C}^{m \times n}$ defines an **invertible** function $A : \mathbb{C}^n \to \mathbb{C}^m$, then from the previous remarks we have $\dim \mathbb{C}^n = \dim \mathbb{C}^m$, hence $n = m$, hence A is square. In conclusion: invertible matrices must be square. If $A \in \mathbb{C}^{n \times n}$, then A is injective ($\mathcal{N}(A) = \{\mathbf{0}\}$) if and only if A is surjective ($\mathcal{C}(A) = \mathbb{C}^n$).

- We define the row space by $\mathcal{R}(A) := \mathcal{C}(A^*) = im(A^*)$. We have $\mathcal{N}(A) = \mathcal{R}(A)^\perp$ inside $\mathbb{C}^n$ and $\mathcal{N}(A^*) = \mathcal{C}(A)^\perp$ inside $\mathbb{C}^m$, hence $\dim \mathcal{R}(A) + \dim \mathcal{N}(A) = n$ and $\dim \mathcal{C}(A) + \dim \mathcal{N}(A^*) = m$. Combining this with the rank-nullity theorem gives
$$\boxed{\dim \mathcal{R}(A) = \dim \mathcal{C}(A)}.$$

- For any matrix A we have
$$\boxed{\mathcal{N}(A^*A) = \mathcal{N}(A)}.$$
Note that A^*A is always square. Then $(A^*A)^{-1}$ exists $\Leftrightarrow$ $\mathcal{N}(A^*A) = \{\mathbf{0}\}$ $\Leftrightarrow$ $\mathcal{N}(A) = \{\mathbf{0}\}$ $\Leftrightarrow$ A has independent columns $\Leftrightarrow$ A has a left inverse. In fact, $L = (A^*A)^{-1}A^*$ is one such left inverse.

- The system $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b} \in \mathcal{C}(A)$. If $\mathbf{b} \notin \mathcal{C}(A)$ then there exists a *least squares solution* $\mathbf{x}$ satisfying $A\mathbf{x} = P_{\mathcal{C}(A)}\mathbf{b}$, where $P_{\mathcal{C}(A)}$ is the orthogonal projection onto the column space. Properties of projection say $A\mathbf{x} - \mathbf{b} \in \mathcal{C}(A)^\perp = \mathcal{N}(A^*)$, hence $A^*(A\mathbf{x} - \mathbf{b}) = \mathbf{0}$, hence $A^*A\mathbf{x} = A^*\mathbf{b}$. If A has independent columns then $\mathbf{x} = (A^*A)^{-1}A^*\mathbf{b}$ is the unique least squares solution to $A\mathbf{x} = \mathbf{b}$.

- Multiply on the left by A to get $P_{\mathcal{C}(A)}\mathbf{b} = A\mathbf{x} = A(A^*A)^{-1}A^*\mathbf{b}$. In other words, if A has independent columns, then
$$\boxed{P_{\mathcal{C}(A)} = A(A^*A)^{-1}A^*}.$$